

ENTHUSIAST ADVANCE COURSE

PHASE : MEC-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 21-05-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	2	1	2	2	2	4	3	4	3	1	3	1	3	2	1	2	1	3	3	3	2	3	2	3	4	2	4	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	4	2	1	1	1	1	1	1	2	1	1	3	2	2	4	3	1	4	2	2	4	1	4	4	4	3	2	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	1	2	3	4	3	3	3	2	2	3	2	3	4	1	4	3	2	4	1	3	3	3	4	1	4	1	1	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	3	2	1	3	3	1	2	3	1	2	3	4	3	2	2	2	2	2	4	3	3	1	3	2	1	4	3	4	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	3	3	2	3	2	2	1	2	2	2	3	4	1	4	3	1	1	4	2	1	3	4	1	3	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	1	4	3	3	3	3	1	4	1	2	3	4	2	3	3	4	4	3	1	3	2	4	2	3	1	4	3	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	3	2	3	3	2	4	1	3	3	2	1	3	2	2	3	1	3	2	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (1)

$$\frac{mg}{m(g-a)} = \frac{3}{2} \quad a = \frac{g}{3}$$

2. Ans (3)

Change in momentum = Impulse

$$= \int F dt$$

= Area of graph

$$= 10 \times 5 + \frac{1}{2} \times 10 \times 4$$

$$= 70 \text{ N s.}$$

3. Ans (2)

$$T \sin \theta = F \Rightarrow T = \frac{F}{\sin \theta}$$

4. Ans (1)

$$a = g \sin \theta - \mu g \cos \theta$$

$$= g[\sin 45^\circ - 0.5 \times \cos 45^\circ]$$

$$= \frac{4.9}{\sqrt{2}} \text{ m/s}^2$$

5. Ans (2)

$$g' = g \left(1 - \frac{d}{R}\right) \Rightarrow \frac{g}{4} = g \left(1 - \frac{d}{R}\right) \Rightarrow d = \frac{3R}{4}$$

6. Ans (2)

$$F_g = F_{CP}$$

$$\frac{GMm}{r^n} = \frac{mv_0^2}{r} \Rightarrow v_0^2 = \frac{GM}{r^{n-1}}$$

$$v_0 = \sqrt{\frac{GM}{r^{n-1}}}$$

7. **Ans (2)**

The escape velocity is independent of mass of the body and the direction of projection. It depends upon the gravitational potential at the point from where the body is launched. Since this potential depends slightly on the latitude and height of the point, the escape velocity depends slightly on these factors.

8. **Ans (4)**

$$V_C = -\frac{3GM}{2R}$$

$$V_r = -\frac{GM}{r} = \frac{V_C}{2}$$

$$\Rightarrow -\frac{GM}{r} = \frac{1}{2} \left(-\frac{3GM}{2R} \right)$$

$$\Rightarrow r = \frac{4R}{3}$$

9. **Ans (3)**

$$\frac{T_2}{T_1} = \left(\frac{r_2}{r_1} \right)^{3/2} = \left(\frac{4R}{R} \right)^{3/2} = 8$$

$$\Rightarrow T_2 = 8T_1 = 8\sqrt{2} \text{ hr.}$$

10. **Ans (4)**

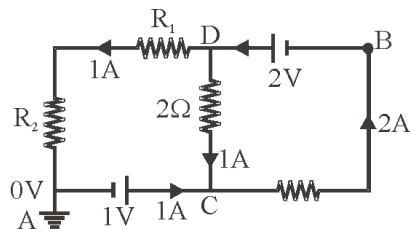
Horizontal force can not balance vertical forces.

11. **Ans (3)**

$$\text{Measured value of } R = \frac{V}{I} = \frac{20}{4} = 5\Omega$$

But current in R will actually be less than I, so R should be greater than 5Ω .

12. **Ans (1)**



By applying KVL along path ACDB
 $V_A + 1 + 2 \times 1 - 2 = V_B \Rightarrow V_B = 1V$

13. **Ans (3)**

When $R_3 = 0$ When $R_3 = \infty$

$$E = I(R_1 + R_2) \quad E = I(R_1 + 2R_2)$$

$$E = 9(R_1 + R_2) \quad E = 6(R_1 + 2R_2)$$

$$9(R_1 + R_2) = 6(R_1 + 2R_2)$$

$$3R_1 = 3R_2 \Rightarrow R_1 = R_2$$

14. **Ans (1)**

T.P.D (V) = E - Ir (Remember it)

$$V = E - \left(\frac{E}{R+r} \right) r = \frac{ER}{(R+r)}$$

from given conditions $E = 2.2$ & when $R = 5$
 then TPD $V = 1.8V$

$$\text{therefore } 1.8 = \frac{2.2 \times 5}{5+r} \Rightarrow r = \frac{10}{9} \Omega$$

15. **Ans (3)**

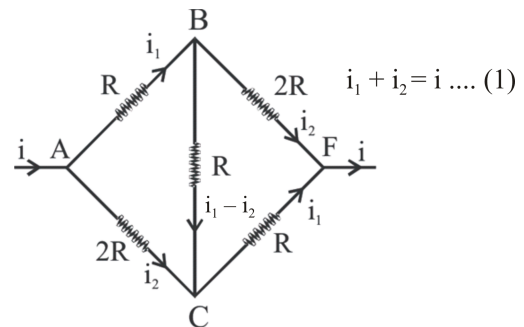
$$J = J_0 \left(\frac{r}{R} \right)$$

$$di = J dA = J (2\pi r dr)$$

$$i = \int_0^R J_0 \left(\frac{r}{R} \right) 2\pi r dr$$

$$i = 2\pi J_0 \left[\frac{R^3}{3R} \right] = \frac{2J_0 A}{3}$$

17. **Ans (1)**



loop ABFA

$$-i_1 R - (i_1 - i_2) R + 2i_2 R = 0$$

$$-2i_1 + 3i_2 = 0$$

$$3i_2 = 2i_1 \quad \dots (2)$$

from (1) & (2)

$$i_1 + \frac{2}{3} i_1 = i$$

$$i_1 = \frac{3i}{5} \text{ Put in (1) } i_2 = \frac{2i}{5}$$

$$\text{Current in arm AD} = i_2 = \frac{2i}{5}$$

18. **Ans (2)**

P.d. across cell $V = E - Ir$

When $I = 0$, $V = E = 2V$

When $V = 0$, $0 = E - 5r \Rightarrow 0 = 2 - 5r$

$$r = 0.4 \Omega$$

19. Ans (1)

Volume remains same $\Rightarrow AL = A'(2L)$

$$A' = ; R_i = \frac{\rho L}{A}$$

$$R_f = \frac{\rho(2L)}{(A/2)} = 4R_i$$

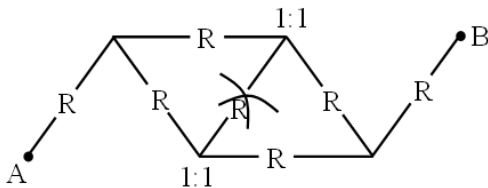
$$\% \text{change} = \frac{4R_i - R_i}{R_i} \times 100 = 300\%$$

20. Ans (3)

Initial all 5R's are in series

$$R_{AB} = 5R \text{ (initial)}$$

Finally



Balanced WSB

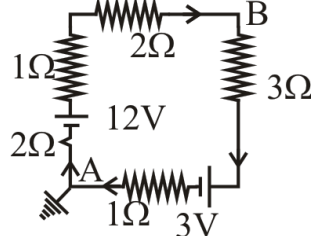
$$R_{AB} = R + R + R = 3R \text{ (final)}$$

so ratio 5 : 3

22. Ans (3)

Current in the circuit

$$I = \frac{12 - 3}{9} = 1A$$



$$V_B - 0 = 3I + 3 + I = 7V$$

Hence option (3)

24. Ans (3)

$$\mu > \frac{1}{(\sin C)}$$

$$\mu > \frac{1}{\sin 45} \Rightarrow \mu > \sqrt{2}$$

27. Ans (4)

$$f = \frac{R}{2(\mu - 1)}, f' = \frac{R}{(\mu - 1)} \Rightarrow f' = 2f$$

29. Ans (4)

$$A_2 (1.72 - 1) = (1.54 - 1)4^\circ$$

$$A_2 = 3^\circ$$

30. Ans (3)

$$\delta = (\mu - 1) A$$

$$34 = (\mu - 1) A \dots(1)$$

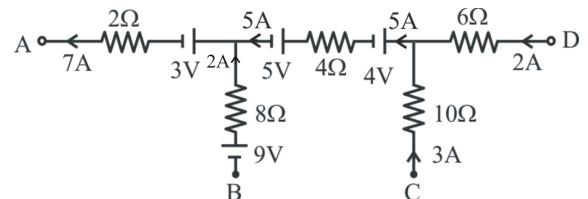
After bnocked off.

$$\delta^1 = (\mu - 1) A/2$$

$$= \frac{34}{2} = 17^\circ$$

SECTION-B

41. Ans (2)



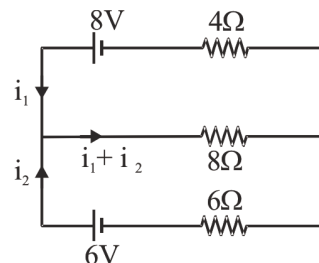
$$\begin{aligned} V_A - V_C &= -3(10) - 4 - 5(4) - 5 - 3 - 7(2) \\ &= -30 - 4 - 20 - 5 - 3 - 14 \\ &= -76 \end{aligned}$$

42. Ans (1)

$$R_2 = \frac{v_2}{0.5} = \frac{20 \times 1}{0.5} = 40\Omega$$

$$R_1 = \frac{69 - v_2}{3.5} = \frac{69 - 20}{3.5} = 14\Omega$$

43. Ans (1)



Use KVL in upper loop

$$8 - 8(i_1 + i_2) - 4i_1 = 0$$

$$\text{or } 8 = 12i_1 + 8i_2 \dots\dots(i)$$

In lower loop

$$6 - 8(i_1 + i_2) - 6i_2 = 0$$

$$6 = 8i_1 + 14i_2 \dots\dots(ii)$$

$$(ii) \times \frac{3}{2} \Rightarrow 9 = 12i_1 + 21i_2 \dots\dots(ii)$$

By (i) & (ii) (ii) - (i)

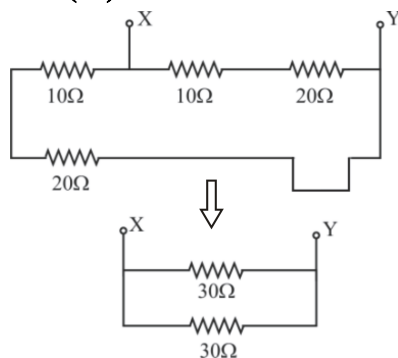
$$1 = 13i_2 \text{ Put in (i)}$$

$$8 = 12i_1 + 8\left(\frac{1}{13}\right) \Rightarrow 2 = 3i_1 + \frac{2}{13}$$

$$3i_1 = \frac{24}{13} \Rightarrow i_1 = \frac{8}{13}$$

$$i_1 + i_2 = \frac{8}{13} + \frac{1}{13} = \frac{9}{13} \approx 0.69 \text{ A}$$

44. Ans (3)



So, $R_{XY} = 15 \Omega$

45. Ans (2)

$$I_1 = I_2 + I_3 \quad \dots(1)$$

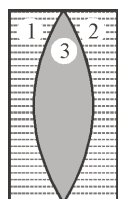
$$\Rightarrow \frac{50 - V}{5} = \frac{V - 0}{3} + \frac{V - 30}{15} \quad \dots(2)$$

$$(2) \times 15 \Rightarrow 3(50 - V) = 5V + V - 30$$

$$150 - 3V = 6V - 30$$

$$\Rightarrow V = 20 \text{ volt}$$

47. Ans (4)



$f_3 = 20$

for plano lens

$R = 20$ each

$$\text{So, } f_1 = f_2 = -\frac{20}{1.6 - 1} = \frac{-100}{3}$$

$$\frac{1}{f_4} = \frac{-6}{100} + \frac{1}{20} = -\frac{1}{100}$$

$$f_{eq} = -100$$

SUBJECT : CHEMISTRY

SECTION-A

65. Ans (4)

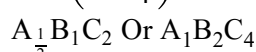
Covalent or network solids are also called giant molecules.

67. Ans (3)

$$A = 4 \times \frac{1}{8} = \frac{1}{2}$$

$$B = 2 \times \frac{1}{2} = 1$$

$$C = \left(4 \times \frac{1}{4}\right) + (1 \times 1) = 2$$



68. Ans (3)

$$(eq.)_{O_2} = (eq.)_{Cu}$$

$$\frac{1.12}{22.4} \times 4 = \frac{w}{63.5} \times 2$$

$$w = 6.35 \text{ g}$$

69. Ans (2)

In equal time interval, equal % of reactant's amount is consumed, so it is a 1st order reaction.

71. Ans (3)

$$\therefore \Delta H = E_a(f) - E_a(b)$$

$$-45 = 30 - E_a(b)$$

$$E_a(b) = 30 + 45 = 75 \text{ K cal}$$

SECTION-B

87. Ans (1)

$$kt = 2.303 \log \frac{a}{a-x}$$

$$\Rightarrow \frac{t_{7/8}}{t_{1/2}} = \frac{\log \frac{1}{1-\frac{7}{8}}}{\log 1 - \frac{1}{2}} = \frac{\log 8}{\log 2} = \frac{3}{1}$$

88. Ans (1)

$$\lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{G^* \times 1000}{R \times N}$$

$$\therefore \lambda_{eq} = \frac{1 \times 1000}{200 \times 0.01} = 500 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

89. Ans (2)

Conductivity of strong electrolytes decreases on dilution.

90. Ans (1)

$$(\lambda_{eq}^\infty)_{Al_2(SO_4)_3} = (\lambda_{eq}^\infty)_{Al^{3+}} + (\lambda_{eq}^\infty)_{SO_4^{2-}}$$

$$= \frac{81}{3} + \frac{90}{2} = 27 + 45 = 72$$

92. Ans (3)

For FCC lattice

$$\sqrt{2}a = 4R \text{ and C.N.} = 12$$

95. Ans (3)

$$\ell n K = \ell n A - \frac{E_a}{RT}$$

SUBJECT : BIOLOGY-I

SECTION-A

101. Ans (2)

NCERT (XIIth) Pg. # 263

SECTION-B

103. **Ans (4)**
NCERT Pg # 264, 265
105. **Ans (2)**
NCERT Pg.# 266
106. **Ans (2)**
NCERT Pg.# 267
108. **Ans (2)**
NCERT XII, Pg. # 271
109. **Ans (2)**
NCERT-XII Page No. 274
110. **Ans (4)**
NCERT (XII) Pg. # 281
111. **Ans (3)**
NCERT (XI) Pg. # 273
112. **Ans (3)**
NCERT Pg. # 283
115. **Ans (2)**
NCERT-XII-Page No. # 25
116. **Ans (1)**
NCERT Pg#33
120. **Ans (2)**
NCERT (XII) Pg. # 25 (E), 26(H)
129. **Ans (2)**
NCERT Pg. # 161, 2st para
130. **Ans (1)**
NCERT Pg. # 175, 3st para

138. **Ans (3)**
NCERT (XII) Pg. # 262, First para
139. **Ans (1)**
NCERT XII, Pg. 281, Fig. 16.7
140. **Ans (1)**
NCERT Pg.# 273
141. **Ans (4)**
NCERT XII pg.# 283
146. **Ans (1)**
NCERT XII, Page # 150, 3rd para
147. **Ans (3)**
NCERT XII, Page # 154, 2nd para
150. **Ans (2)**
NCERT (XIIth) Pg. # 146 Last para

SUBJECT : BIOLOGY-II
SECTION-B

188. **Ans (1)**
NCERT-XII, Pg. # 151
189. **Ans (3)**
NCERT-XII, Pg. # 159
190. **Ans (3)**
NCERT-XII, Pg. # 155
191. **Ans (2)**
NCERT XII, Page # 151
198. **Ans (3)**
NCERT- (XII) Pg. # 149